Dr. SUI Xiangjie

Personal Information

Assistant Professor
Faculty of Data Science
City University of Macau
Macau SAR, China

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Google Scholar: https://scholar.google.com/citations?user=IAnTG2cAAAAJ&hl

Interests: Computer Vision, Image/Video Processing, Deepfake Detection, Visual Quality Assessment, Model Robustness, Virtual Reality

Education

09/2021–06/2024 **Ph.D., Service Computing and Applications**
Jiangxi University of Finance and Economics, Nanchang, China
Thesis: Perceptual quality assessment of panoramic images based on visual scanpath
Advisor: Prof. Fang Yuming

09/2018–06/2021 **M.Sc., Computer Science and Technology**
Jiangxi University of Finance and Economics, Nanchang, China
Thesis: Perceptual quality assessment of asymmetrically distorted 3D videos
Advisor: Prof. Fang Yuming

09/2014–06/2018 **B.Sc., Computer Science and Technology**
Jiangxi University of Finance and Economics, Nanchang, China

Working Experience

08/2024–Now **Faculty of Data Science, City University of Macau**
Position: Assistant Professor

05/2023–06/2024 **Dept. of Computer Science, City University of Hongkong**
Position: Research Assistant
Supervisor: Prof. Wang Shiqi

Teaching

Database System, spring, 2025, 2026.

Linear Algebra, fall, 2024, 2025.

Introduction to Operating Systems, fall, 2024, 2026.

Honors, Awards & Scholarships

Best Poster Paper, AAAI Conference on Artificial Intelligence (AAAI), 2026.

Excellent Master's Thesis in Jiangxi Province, China, 2023.

Outstanding Report Award of the 2nd Postgraduate Academic Forum of Journal of Image and Graphics, 2022.

National Scholarship for Doctoral Student, 2022.

China Council Scholarship, 2021.

Top-15 most prominent papers published in Journal of Image and Graphics, 2021.

National Scholarship for Master Student, 2020.

Projects

2026	3D 可信身份無感認證技術及其應用研究 Role: **Project Member**; Funding: **FDCT**
2025	面向服務機器人的具身多模態大模型關鍵技術研究 Role: **Project Member**; Funding: **FDCT-MOST**, No. 0007/2025/AMJ
2025	基於空天遙感的城市智能感知關鍵技術研究 Role: **Project Member**; Funding: **FDCT**, No. 0001/2025/AIJ

Community Services

Journal Reviewer

IEEE Transactions on Image Processing (TIP); IEEE Transactions on Multimedia (TMM); IEEE Transactions on Circuits and Systems for Video Technology (TCSVT); IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI); IEEE Signal Processing Letters (SPL)

Conference Reviewer

Conference on Neural Information Processing Systems (NeurIPS); IEEE International Conference on Computer Vision (ICCV); IEEE Conference on Computer Vision and Pattern Recognition (CVPR); International Conference on Machine Learning (ICML, ***Gold Reviewer, 2026***); European Conference on Computer Vision (ECCV); IEEE International Conference on Multimedia and Expo (ICME); British Machine Vision Conference (BMVC)

Supervisions

Supervised MSc Students

2025	**Qiu Zhaotian**	**Thesis**: Visual attention modeling
	Wu Haocheng	**Thesis**: Visual quality assessment
2024	**Li Songyang**	**Thesis**: VLM robustness
	Su Yanguang	**Thesis**: Panoramic video cinematography

Supervised BSc Students

2025	**Li Zixuan**	**Thesis**: Panoramic video cinematography
	Xu Jinghao	**Thesis**: Intelligent vocabulary learning system
2024	**Zhang Xuanwei, Li Yitian, Nie Junquan, Lin Zenan** **Thesis**: Automatic cinematography for panoramic videos	

Publications

Preprints

[P1] **Sui, Xiangjie**, Li, S., Zhu, H., Chen, B., Fang, Y., and Sun, X. Diagnosing corruption-induced reliability failures in vision-language models. *arXiv preprint arXiv:2511.19032* (2025)

Journal Publications

[J1] **Sui, Xiangjie**, Zhu, H., Liu, X., Fang, Y., Wang, S., and Wang, Z. Perceptual quality assessment of 360° images based on generative scanpath representation. *IEEE Transactions on Image Processing 34* (2025), 4485–4499

[J2] Zhu, H., **Sui, Xiangjie***, Chen, B., Liu, X., Chen, P., Fang, Y., and Wang, S. 2afc prompting of large multimodal models for image quality assessment. *IEEE Transactions on Circuits and Systems for Video Technology 34*, 12 (2024), 12873–12878

[J3] Wen, W., Li, M., Yao, Y., **Sui, Xiangjie**, Zhang, Y., Lan, L., Fang, Y., and Ma, K. Perceptual quality assessment of virtual reality videos in the wild. *IEEE Transactions on Circuits and Systems for Video Technology 34*, 9 (2024), 8368–8381

[J4] Fang, Y., Li, Z., Yan, J., **Sui, Xiangjie**, and Liu, H. Study of spatio-temporal modeling in video quality assessment. *IEEE Transactions on Image Processing 32* (2023), 2693–2702

[J5] 鄢杰斌, 方玉明, 刘学林, 姚怡茹, and **眭相杰**. 视频质量评价研究综述. 计算机学报 *46*, 10 (2023), 2196–2224

[J6] **Sui, Xiangjie**, Ma, K., Yao, Y., and Fang, Y. Perceptual quality assessment of omnidirectional images as moving camera videos. *IEEE Transactions on Visualization and Computer Graphics 28*, 8 (2022), 3022–3034

[J7] Fang, Y., **Sui, Xiangjie**, Wang, J., Yan, J., Lei, J., and Le Callet, P. Perceptual quality assessment for asymmetrically distorted stereoscopic video by temporal binocular rivalry. *IEEE Transactions on Circuits and Systems for Video Technology 31*, 8 (2021), 3010–3024

[J8] 方玉明, **眭相杰**, 鄢杰斌, 刘学林, and 黄丽萍. 无参考图像质量评价研究进展. 中国图象图形学报 *26*, 2 (2021), 265–286

[J9] **Sui, Xiangjie**, Ding, M., Yan, J., Fang, Y., Zuo, Y., and Tan, Z. Objective quality assessment of synthesized images by local variation measurement. *Signal Processing: Image Communication 92* (2021), 116096

[J10] Fang, Y., **Sui, Xiangjie**, Yan, J., Zuo, Y., Wang, J., and Li, Z. Asymmetrically distorted 3d video quality assessment: From the motion variation to perceived quality. *Signal Processing 183* (2021), 108031

Conference Publications

[C1] Chen, B., Pan, S., Wu, D., Xie, L., **Sui, Xiangjie**, Zhu, L., and Zhu, H. Mitigating perception bias: A training-free approach to enhance LMM for image quality assessment. In *Proceedings of the AAAI Conference on Artificial Intelligence* (2026), vol. 40, pp. 2760–2768. **(Best Poster Paper)**

[C2] Zhao, Z., Wang, M., **Sui, Xiangjie**, He, X., and Wang, S. CTU-level rate control with λ optimization based on visual gaze mechanism for 360-degree versatile video coding. In *IEEE International Conference on Image Processing* (Sep. 2025), pp. 1–6

[C3] **Sui, Xiangjie**, Wang, S., and Fang, Y. A survey on objective quality assessment of omnidirectional images. In *Asia Pacific Signal and Information Processing Association Annual Summit and Conference* (Dec. 2024), pp. 1–6

[C4] **Sui, Xiangjie**, Fang, Y., Zhu, H., Wang, S., and Wang, Z. Scandmm: A deep markov model of scanpath prediction for 360deg images. In *IEEE Conference on Computer Vision and Pattern Recognition* (June 2023), pp. 6989–6999

[C5] Fang, Y., Yao, Y., **Sui, Xiangjie**, and Ma, K. Subjective quality assessment of user-generated 360° videos. In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops* (Mar. 2023), pp. 723–724

[C6] Li, G., **Sui, Xiangjie**, Yan, J., and Fang, Y. Benchmarking 360° saliency models by general-purpose metrics. In *IEEE International Workshop on Multimedia Signal Processing* (Sep. 2022), pp. 1–6

[C7] Fang, Y., **Sui, Xiangjie**, and Wang, J. A spatial-temporal weighted method for asymmetrically distorted stereo video quality assessment. In *IEEE International Symposium on Circuits and Systems* (May 2019), pp. 1–5

July 29, 2026